

Investigating Experimental Design+Theater+Coding Methodology to Teach Virtual Reality Design: A Case Study

Investigating Experimental Design+Theater+Coding Methodology

Daria Tsoupikova
CS+DES, School of Design, University
of Illinois Chicago
datsoupi@gmail.com

Jeff Nyhoff
Computer Science, Trinity Christian
College
jnyhoff@gmail.com

Karla Rangel
IBM, Supply Chain
krangelg@mx1.ibm.com

ABSTRACT

This talk presents a case study of how two Chicago-area faculty based at two different academic institutions collaboratively investigated the use of an original, experimental, and cross-disciplinary “Design+Theater+Coding” methodology for the teaching and guiding of University of Illinois Chicago (UIC) undergraduate design students in the creation of thought-provoking and empathy-engaging interactive Virtual Reality team projects regarding global issues, all within the context of a single-semester creative coding course in Fall 2021. This experiment explored and leveraged the relationship between theater, creative coding, and VR technology and considered how the relationship might enhance design studies.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Virtual reality, Interdisciplinary collaboration, CAVE2TM, Education, Design, Creative coding, Theater

ACM Reference Format:

Daria Tsoupikova, Jeff Nyhoff, and Karla Rangel. 2022. Investigating Experimental Design+Theater+Coding Methodology to Teach Virtual Reality Design: A Case Study: Investigating Experimental Design+Theater+Coding Methodology. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '22 Educator's Forum)*, August 07–11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532724.3535595>

1 INTRODUCTION

Design and Theater methodologies have influenced Creative Coding and software development for more than a half-century. In the 1970s, when Alan Kay and his Xerox PARC team looked to the idea of theatre when they designed the revolutionary Dynabook personal computer and Smalltalk programming environment that first showcased the immersive graphical user interface experiences we common in the Mac, Windows, and touchscreen devices used widely today. [Kay, 2002]

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Educator's Forum, August 07–11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9366-9/22/08.

<https://doi.org/10.1145/3532724.3535595>

Brenda Laurel's highly influential 1991 book, *Computers as Theatre*, drew parallels between immersive end-user interface experiences and theatrical performance, offering the idea of theatre as both an explanation and a strategy for the design of user interaction experiences to an industry where, problematically, “a scientific approach dominates the design of human-computer interaction.” [Laurel, 1991] In retrospect, this should not seem surprising: both user experience (UX) and theatre rely heavily upon shared action, character, language, space, empathy, and improvisation to create thoughtful and emotionally engaging communication between hearts, minds, and bodies, during both the design process and the resulting experience.

However, in spite of the historical and contemporary realities of the mutual relevance and potential enrichment that the theories, practices, vocabularies and methodologies of theatre, design, and technology stand to offer each other, they remain largely distant and disjoint academic disciplines, discouraging collaboration and confluence between these specialized discourses and practices.

In contrast, this project arose from the shared and complementary interdisciplinary interests and experiences of two faculty who are situated at separated academic institutions and in different academic disciplines. Online conversations on the potential relationship of user software design to the concepts and practices of theatre, performance, and improvisation that were convened by Karla Rangel and Jeff Nyhoff engaging global IBM meetings in 2021. Tsoupikova, the Director of Computer Science + Design (CS+DES) new interdisciplinary program subsequently joined. Because of the success of these conversations, the three decided to infuse these theatricalities into the methodological approach to the Fall 2021 Creative Coding Course that Tsoupikova and Nyhoff would be co-teaching at UIC, with IBM supporting this experiment.

2 DESIGN+THEATER+CODING APPROACH

Our course empowered student teams to explore the VR CAVE2 as a medium for the design of an experience that is both virtual and embodied, immersing the interactant in an encounter with such pressing and problematic present-day human experiences as water pollution, air pollution, migration, poverty, inclusiveness and diversity, and mental health. [Febretti et al., 2013]

Student designers from UIC's Industrial Design and Graphic Design programs collaborated on seven experimental VR projects, working within the design constraint of typographic expressions along a steep, eight-week learning curve. Their VR projects were developed by employing a considerable array of software tools that included 3D modeling packages for the Unity game engine,



Figure 1: Student VR project “Barriers” by John Lee, Agustín Tena, Abril Azpeitia during the review in the CAVE2 in the Electronic Visualization Laboratory (EVL). (Photo: John Lee, Agustín Tena, Abril Azpeitia)

the Adobe Creative Suite, Maya, Audacity, and collaboration tools for virtual teamwork. They wrote and customized Unity scripts using the C# programming language, even though almost all of the students were new to coding. Similarly, most of the students had little or no prior experience in theatre but, throughout the design process, were asked to consider and to explore how the vocabularies, concepts, methodologies, and technologies of coding, theatre, and design could complementarily inform and invigorate their work of creating a thought-provoking and empathy-engaging VR interactant experience related to their chosen topic. Along the way, the instructors and students collaboratively explored such key questions as the following:

What might creative coding, VR, design, and theater have in common, conceptually and practically?

- Can the VR experience be designed in such a way that it raises our awareness of the hidden systems underlying both technologies and global environmental challenges and urges us toward a critique and interrogation of them?
- Can creative coding adapt theater and design concepts and methodologies to create VR interactant experiences that are not only engaging but also edifying?

In this course, we investigated an experimental approach to design VR concepts based on intersections of the vocabularies, theories, and practices of theatre, performance, and creative coding. For example, we drew parallels between the audience and user experience. We explored about how theatre’s prioritization of the construction and representation of action can structural strengthen the interactive experience in VR. We considered how the empathic and motivational relationship the character that many stage actors seek in the theatre might enable the design of VR experiences that are more effective and nuanced in the evoking empathy from the interactant in regard to the societal concerns that were the topics of this course.

Class sessions included demonstrations, presentations, discussions, hands-on work, workshops, and teamwork. Students also explored the history of the VR art and design, creative coding, real-time typography, theater, and ethics.

The final projects were exhibited in the CAVE2 (Figure 1) and later in an online gallery. This final exhibition was framed within a discussion of how design and theater can help mitigate issues related to global cultural, health, and environmental concepts impending our world.

3 DISCUSSION

Our talk will present takeaways about effective new teaching techniques such as supportive online resources, evaluation of collaborative contributions, and supporting materials. We will share our insights in relation to UIC’s new interdisciplinary CS+DES curriculum. In addition, we will share how theater methodologies, creative coding, and VR technology can complementarily enhance a design curriculum. We believe that the approach we are undertaking both supports and encourages student creativity, inspires experimental design and collaboration, and enhances the overall learning experience.

REFERENCES

- Kay A. Keynote address to Digital Cultures Project, University of California at Santa Barbara. 2002.
- Laurel B. *Computers as Theatre*. Addison-Wesley; 1991. xv.
- Febretti A., Nishimoto A., Thigpen T., Talandis J., Long L., Pirtle JD., Peterka T., Verlo A., Brown M., Plepys D., Sandin D., Renambot L., Johnson A., Leigh J., CAVE2: A Hybrid Reality Environment for Immersive Simulation and Information Analysis. *Proc. SPIE 8649, The Engineering Reality of Virtual Reality 2013*, 864903 (2013)